Set-A

BRAC UNIVERSITY Inspiring Excellence	**BRAC University** **Department of Computer Science and Engineering** Semester: **Summer 2025** Course Code: **CSE460** Course Title: **VLSI Design**	**Midterm Examination** Full Marks: **10 x 2 + 15 = 35** Time: **1 hour 20 minutes** Date: **26 July 2025**

Student ID:	**Name:**	**Section:**

[Answer all **THREE** questions. Return the question paper with the answer script.]

Question 1: [CO2]

Kelvin is tasked with building an FSM which would detect **10 or 011 overlapping** sequences coming out of a communication channel. A sample is given below:
(w indicates input sequence values and z indicates output)

Clock cycle	t0	t1	t2	t3	t4	t5	t6	t7	t8	t9	t10	t11	t12	t13
w	1	0	1	1	0	0	1	1	0	1	0	1	0	0
z	0	0	**1**	0	**1**	**1**	0	0	**1**	**1**	0	**1**	0	**1**

Note: Use the table below for **Gray coding** if needed.

Decimal Equivalent	**Binary Code**			**Gray Code**		
	B2	**B1**	**B0**	**G2**	**G1**	**G0**
0	0	0	0	0	0	0
1	0	0	1	0	0	1
2	0	1	0	0	1	1
3	0	1	1	0	1	0
4	1	0	0	1	1	0
5	1	0	1	1	1	1
6	1	1	0	1	0	1
7	1	1	1	1	0	0

(a)	**Determine** the type of **FSM** (Mealy/Moore) to be implemented	[1]
(b)	**Draw** the state diagram. Use no more than **6 states**. Using more states will result in deduction of marks.	[4]
(c)	**Draw** the state assignment table using **Binary encoding**.	[3]
(d)	**Determine** the expression of only the output, **z**. No need to draw circuit.	[2]

Question 2: [CO2]

Design a finite state machine (FSM) for a car alarm system. The system has two binary inputs: x, indicating the lock status of the car door (**x = 1 if unlocked, x = 0 if locked**) and y, indicating the door status (**y = 1 if the door is open, y = 0 if closed**). The output z is used to control the alarm: z = 1 when the alarm is active, and z = 0 otherwise. The alarm should be triggered (z = 1) if the door is opened (y = 1) while it is still locked (x = 0) **for two consecutive clock cycles**. An example is shown below:

x	1	0	1	0	0	0	1	1	0	0	0	0	1	0
y	0	0	1	1	1	0	1	0	1	1	1	0	0	1
z	0	0	0	0	**1**	0	0	0	0	**1**	**1**	0	0	0

(a)	**Determine** the type of **FSM** that would be needed for the given problem and design the state diagram for the corresponding FSM.	[2]
(b)	**Derive** the state-assigned table using the **Gray encoding** from your state diagram in (a).	[2]
(c)	**Find** the expression for the next state and the output for the FSM using **KMAP**.	[3]
(d)	**Calculate** the number of flip-flops required for the circuit implementation and **draw** the circuit for this FSM.	[3]

Question 3: [CO1]

Scenario 1:

A classroom has **three** temperature sensors to control the temperature of the room by controlling the air conditioner installed in that room. The temperature sensors produce a high output if the temperature crosses a certain threshold. The air conditioner turns **on** if the outputs of all three sensors are **high**, otherwise it remains **off**.

Scenario 2:

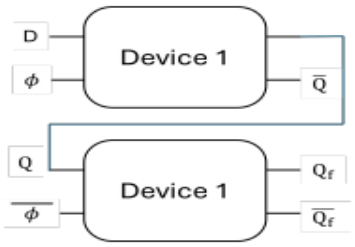

Figure: 1

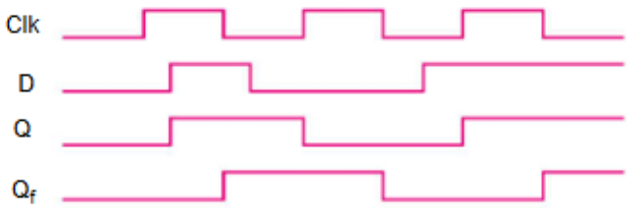

Figure: 2

Figure 2 shows the output waveshape of the device implemented in **Figure 1**.

(a)	**Identify** the logic circuit required to implement the system mentioned in scenario 1 and implement the circuit using **2:1 mux**.	[5]
(b)	**Identify** the circuit implemented in **Figure 1** from the timing diagram of **Figure 2**. Also identify **Device 1** shown in **Figure 1**.	[3]
(c)	**Draw** the circuit of **Device 1** using transmission gates.	[3]
(d)	**Explain** why **n-MOS** pass transistors produce good 0 and bad 1, and **p-MOS** pass transistors produce bad 0 and good 1 with proper mathematical justification.	[4]